Project Record - part 8 of 8

Utensil Theories Transfer Protocol, 17 to 21 October 2026. This part holds: 11. How to reproduce the work; 12. Plain-language glossary; 13. Things outside the AI's control.

11. How to reproduce the work

- **Versions used:** Python 3.13.16, torch 2.14.1, numpy 2.5.3 (scipy 1.18.1 for experiments 1-4), on 4 CPU cores. Reproducing a seed exactly needs the same versions.
- **Threads:** training uses 4, the long-story test 1, exact seeds 2. Exact regrowth needs 2 threads; a 1-thread decoder fails.
- **Downloads:** only GitHub-hosted files could be fetched in the cloud machine; other book sites were blocked.

Experiments 1 to 11 (from `experiments/HANDOFF.md`) and experiment 12 (from the `GROW_UNDERSTANDING_2026-10-20` guide):

```
pip install numpy scipy torch
cd experiments
python bible_understanding.py > results.txt              # exp 1-4
python consequences_test.py > results_consequences.txt   # exp 5
python tiny_transformer.py --only full --seeds 0         # exp 6
python shifting_word.py --seed 0                         # exp 7
python confirm.py                                        # exp 8
python method_comparison.py                              # exp 10
python long_story_test.py                                # exp 11
python aesop_test.py                                     # exp 9
python understanding_compression.py                      # exp 12, about 24 min
```

- `confirm.py`, `method_comparison.py`, `long_story_test.py` and `aesop_test.py` re-print their analysis from the committed results without retraining.
- To regrow seeds, unzip the three model zips (or use `experiments/models/`), then check each model with `sha256sum -c SHA256SUMS.txt` inside the models folder.

The seed test (any computer with Python 3, no extra packages):

```
cd SEED_TEST_2026-10-20
python seed_test.py
python seed_test.py --method graph_fit photo.jpg letter.txt
```

The numbers test (from `experiments/`):

```
python numbers_test.py --dry-run --scratch /some/scratch/folder
python numbers_test.py               # full run, 18 models, resumable
python numbers_test.py --report-only  # rebuild tables from results_numbers.json
bash NUMBERS_TEST_2026-10-21/code/run_full.sh   # from the top level: full run plus 10-minute auto-save
```

12. Plain-language glossary

- **LLM (large language model):** a large AI that reads and writes text, such as Claude, Gemini or Copilot.
- **Tiny model / tiny transformer:** the small model trained from scratch here: 3.4 million parameters, 4 layers.
- **Parameter:** one adjustable number inside a model. Training tunes them.
- **Transformer:** the model design behind modern AIs; it predicts the next token from the tokens before it.
- **Token:** one piece of text the model reads, here a whole word or a punctuation mark.
- **Tokenizer:** the rule that cuts text into tokens. Here `[a-z]+|[.,;:?!]`, which drops digits.
- **Sub-word tokenizer:** cuts words into shared pieces, so related spellings share parts. Not yet tried.
- **Context window:** how far back the model can see: 64 tokens here (128 in the Aesop test).
- **Training step:** one small update of the model. Most models here had 1,500 steps (the two small Aesop models about 275; dry-run models 5).
- **Seed (random start):** the number that fixes a model's random starting point. Seeds 0, 1, 2 are three independent runs.

- **Seed (data seed):** the owner's idea: a tiny message that regrows exact data using shared grow instructions.
- **Greenhouse:** the owner's word for the shared grow instructions, such as a trained model both sides hold.
- **Embedding / word vector:** a list of numbers standing for a word's meaning, learned from which words appear together.
- **PPMI:** a score for how much more often two words appear together than chance would give.
- **SVD:** a method that squeezes a large table of numbers down to its main directions (100 here).
- **Held-out:** text kept out of training and used only for testing. Here every 50th verse.
- **Control:** a version identical except for the idea being tested.
- **Scrambled control:** the shifting word with spellings chosen at random: same extra word, no meaning.
- **Shuffled stories:** stories with sentences shuffled; same words, broken event order.
- **Baseline:** the plain model with no marker.
- **Plain label:** the established method: each verse marked good or bad on its own, no carry.
- **Shifting word:** the owner's made-up word whose spelling shows a running balance.
- **Remaining:** what is left of the balance after a verse; it carries to the next appearance.
- **Reset:** a clear right or wrong sets the balance to +17 or -17.
- **Clear choice:** a verse that triggers a reset (verdict phrases, or the 3% most strongly worded at each end).
- **Regex:** a text-matching pattern, used to find verdict phrases.
- **Spend:** the old rule: each valued word moves the balance 0.5 toward zero.
- **Drain:** the numbers-test rule: each valued word shrinks the balance by 14.1%.
- **Recharge:** re-adding a charge (17, or the owner's numbers) each time the word appears.
- **Cap:** the limit on the balance, +/-51. "Pinned" means stuck at the cap.
- **Manipulation check:** a check before training that a rule really changes the text (here: no more than 5% at the cap; every spelling on at least 3% of verses).
- **Dry run:** a fast run of every step with tiny settings, to test the machinery only.
- **Pre-registration:** pass/fail rules saved to git before any result is seen.
- **Chance / coin toss:** 50% right by guessing between two options.
- **Tie:** the model scores both stories exactly the same; counted as half right.
- **Greedy marker choice:** the model always writes its single most likely spelling; may hide small updates.
- **Sign test:** counts pairs where one method won and the other lost, and asks if that split could be luck.
- **Permutation test:** reshuffles the labels many times (100,000 here) to see how often luck matches the real gap.
- **p-value:** the chance of a result at least this strong if there were no real effect. Below 0.05 counts as a pass.
- **Holm correction:** a stricter bar when several tests are run at once, so one lucky pass is not over-read.
- **Confirmed / supported once / not confirmed / untested / running:** passed a pre-registered test / passed once, after a fix or without a fresh repeat / did not pass / not yet tried / started, with no results yet.
- **Arithmetic coding:** turns a model's predictions into bits; good predictions mean fewer bits.
- **Bit:** one 0 or 1. "Bits per character" measures how tightly text is packed.
- **Entropy:** the true amount of surprise in data; no exact method can go below it on average.
- **Kolmogorov complexity:** the length of the shortest program that regrows a file. It cannot be computed in general.
- **Pigeonhole / counting limit:** there are fewer short seeds than files, so no method shrinks every file, and random data never shrinks.
- **SHA-256 fingerprint:** a 64-character code computed from a file; any change gives a different code.
- **Byte for byte / exact regrowth:** the regrown file is identical, checked by fingerprint.
- **Thread:** one of the processor's lanes of work. A different thread count can change tiny rounding and break exact regrowth.
- **zip / xz:** standard compression tools, used for comparison.
- **Drift / drift guard:** sliding from the goal toward the measuring tool; the guard is the written check against it.

13. Things outside the AI's control

- **The repository is public** on GitHub, with an MIT license: anyone can read and reuse everything in it, including this record. Some zine-project files there contain the owner's contact details. Changing visibility is the owner's decision, in GitHub settings. If a patent is planned, ask a patent attorney before more is made public.
- **The cloud machine is temporary.** Anything not committed and pushed is lost when it is deleted. The Aesop models (235 MB) live only on this machine.
- **Files cannot be placed on the owner's devices.** A cloud session cannot put files on a laptop desktop or phone home screen, or make widgets. Files are sent in the app and by email for the owner to save.
- **Scale honesty.** Every result comes from tiny models trained on one book. None of it changes how a large AI such as Claude understands anyone. Nothing has been tested on a large AI or in a real conversation; that needs an API key the owner would add.
- **Usage limits and safety checks.** Session limits stopped work twice on 20 Oct and twice on 21 Oct. On 21 Oct between 01:42 and 01:53 UTC, an automatic safety check blocked 5 replies to the owner's review request, with no reason shown. The AI cannot see or change why, and it has no access to billing or usage refunds (those go to Anthropic support).
- **Privacy in this record.** Some of the owner's messages and uploaded documents contain personal details and other people's names. Because this record is public, those are summarized in one neutral line or left out, never quoted. The owner still has the originals.